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## What is a Gantt chart?

A Gantt chart is a project management tool assisting in the planning and scheduling of projects of all sizes; they are particularly useful for visualising projects. A Gantt chart is defined as a graphical representation of activity against time; it helps project professionals monitor progress.

Gantt charts are essentially task scheduling tools: project management timelines and tasks are converted into horizontal bars (also called Gantt bars) to form a bar chart. These

Gantt bars show start and end dates, as well as dependencies, scheduling and deadlines,



including how much of the task is completed per stage and who is the task owner. Gantt charts show planned activity against time; they are frequently used throughout projects, programmes and portfolios after tasks have been identified using a work breakdown structure.

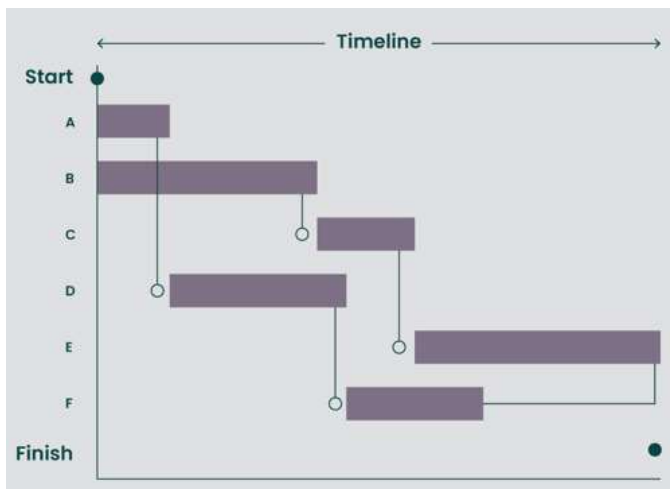
A timeline, like the Gantt chart is useful to keep tasks on track when there is a large team and multiple **stakeholders**. They are a useful time management and progress tracking tool – you can also use Gantt charts to find the longest path from project start to project completion which is known as the **critical path**.

As it's a bar chart format, it's possible to check progress with a quick glance. You can easily see:

- a visual display of the whole project
- timelines and deadlines of all tasks
- relationships and dependencies between the various activities
- project phases

Project management solutions that integrate Gantt charts give project managers insights into team workloads, as well as current and future availability, which allows for more accurate scheduling. Gantt charts have been around for nearly a century, having been invented by Henry Gantt, an American mechanical engineer, around 1910.

## How to use a Gantt chart



The underlying concept of a Gantt chart is to map out the tasks to be completed and

their order. This can identify tasks that can be done in parallel – at the same time, or in sequence – one after another. If we combine this with the project resources we can explore the trade-off between the scope (doing what needs to be achieved), cost (using more or less resources) and the time scales for the project. By adding more resources or changing the scope the project manager can see the effect on the end date.

To create a Gantt chart you need to know all of the individual tasks required to complete the project, an estimate of how long each task will take and which tasks are dependent on others. The very process of pulling this information together helps a project manager focus on the essential parts of the project and begin to establish a realistic timeframe for completion.

**In summary:**

- When you set up a Gantt chart, you need to think through all the tasks involved in your project and divide them into manageable components.
- Then decide who will be responsible for each task, Gantt charts support resource allocation.
- Identify task relationships and decide on the completion date sequence for each task, showing the expected time duration of the whole project and the sub tasks. A Gantt chart will show the tasks in a sequential order and display task dependencies (i.e. how one task relates to another).
  - Decide how long each task is likely to take.
  - Decide the order in which tasks need to be completed.

- Identify areas where you don't have enough resources (too many activities being allocated to one person), or tasks that will taking too long to complete (the order of activities results in the project being completed later than anticipated).
  - Identify areas where the schedule can be refined, for example, can the date of any task change? Can you get more resources?
- Determine and allocate your resources.
- Anticipate the risks and problems you may encounter and create a contingency plan for potential problems.

# What are the pros and cons of a Gantt chart?

Advantages of Gantt charts:

- clear, visual representation of the plan
- generally well understood and relatively simple to create
- can be used to show progress and plan resources.

Disadvantages of Gantt charts:

- on their own, they don't show dependencies
- there is a limit to the size of schedule that can reasonably be read and understood.
- cannot easily cope with change as a result of progress or scope change

## Planning, scheduling, monitoring and control

Planning, Scheduling, Monitoring and Control offers practical guidance on all planning aspects of preparing to undertake a project, executing a project, controlling its delivery to budget, time and quality, and delivering it safely.

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## Blog: History of the Gantt chart

by Paul Naybour

It is difficult to envisage how a modern project would be managed without at some point creating a chart of tasks to be done in delivering the project's declared benefits.

One of the most enduring types of chart is the Gantt chart.

[Read blog](#)

## How to make a Gantt chart for projects in Excel

Microsoft Excel has a bar chart feature that can be formatted to make an Excel Gantt chart.

## Gantt chart software

Here are some popular Gantt chart tools:

- [Primavera P6](#)
- [Apache OpenOffice](#)
- [Microsoft Project](#)
- [ProjectLibre](#)
- [OpenProj](#)
- [Microsoft Excel project planner](#)

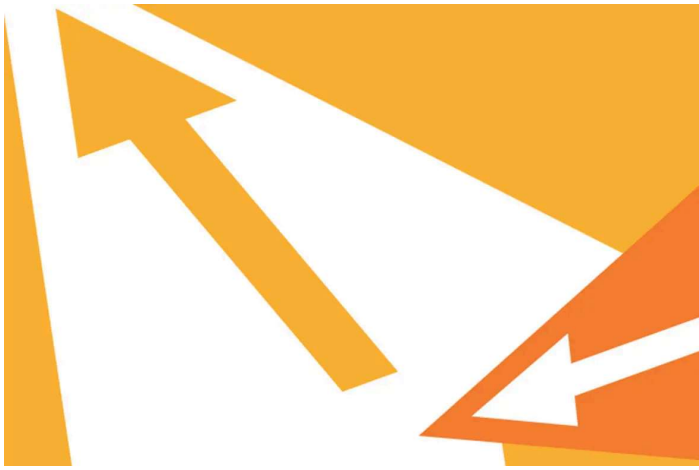
- **Step 1.** List your project schedule in an Excel table. List each task as a separate row and structure your project plan by including the start date, end date and duration.
- **Step 2.** Set it up as a standard bar chart based on start dates.
- **Step 3.** Add the duration of your tasks.
- **Step 4.** Add task descriptions to the Gantt chart.  
Sometimes Gantt charts also display the cost and budget data for each activity; and progress data such as actual dates, percentage complete and remaining duration.
- **Step 5.** Transform the bar graph into the Excel Gantt chart through formatting.

For step-by-step instructions see [the full article](#). You can also do this in PowerPoint, suitable for presentations. Microsoft has an [Excel Gantt chart](#) template which you can download.

## Starting out in project management

Your essential guide to the basics of project management. Written for anyone new to projects or wishing to progress their career as a project professional *Starting Out* charts the journey of the APM project life cycle, from concept through to delivery and handover.

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## Related reading

[Planning beyond the Gantt chart](#)

[Managing flexibility in your project schedule](#)

[What is schedule management?](#)

[Is the project Gantt chart still relevant today?](#)

[Using a Gantt chart to manage a project schedule](#)

[The basics of an effective project plan](#)



## What is project management?

Project management is the application of processes, methods, knowledge, skills and experience to achieve the project objectives.

Projects are the vehicles of change in an organisation. They are not the repetitive business-as-usual type of activity.

What is project management?

## What is agile project management?

Agile project management is an approach based on delivering requirements iteratively and incrementally throughout the project life cycle.

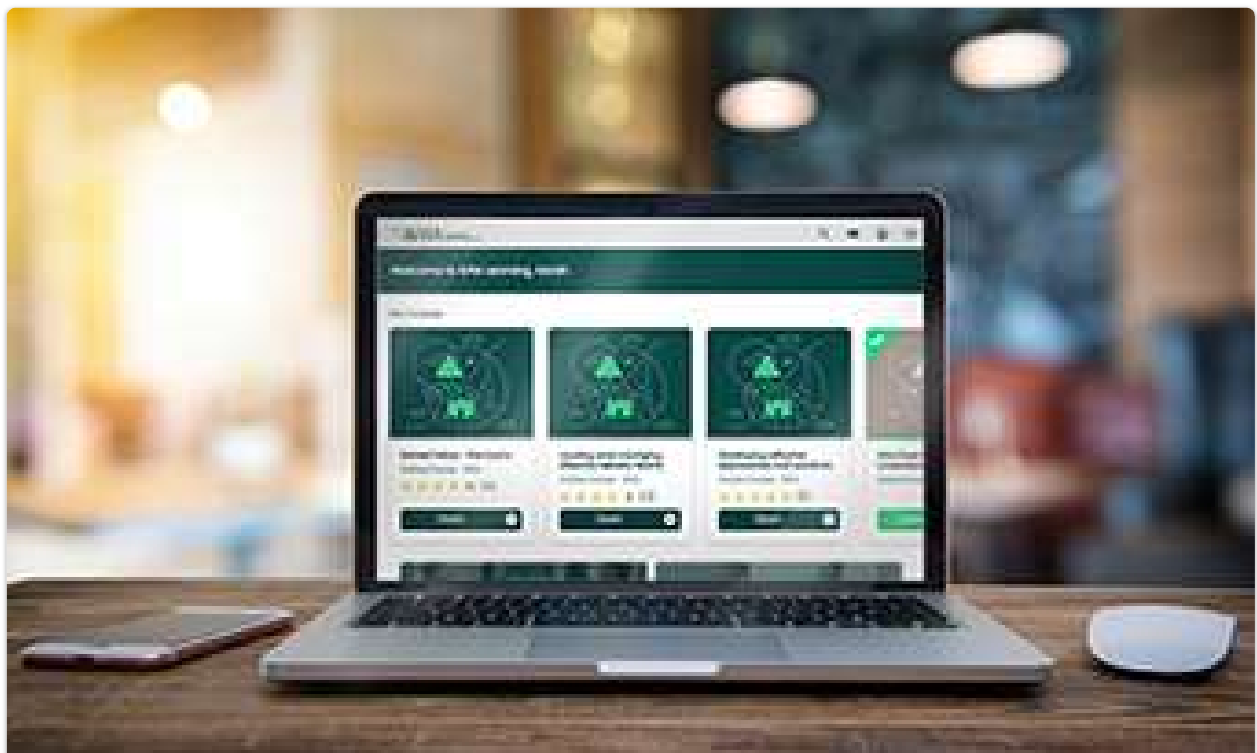
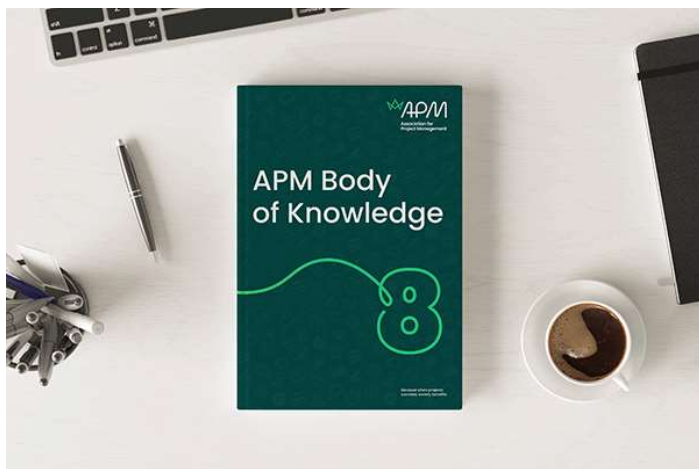
At the core of agile is the requirement to exhibit central values and behaviours of trust, flexibility, empowerment and collaboration

What is agile project management?

## ***APM Body of Knowledge 8<sup>th</sup> edition***

The *APM Body of Knowledge* is a foundational resource providing the concepts, functions and activities that make up professional project management. It reflects the developing profession, recognising project-based working at all levels, and across all sectors for influencers, decision makers, project professionals and their teams.

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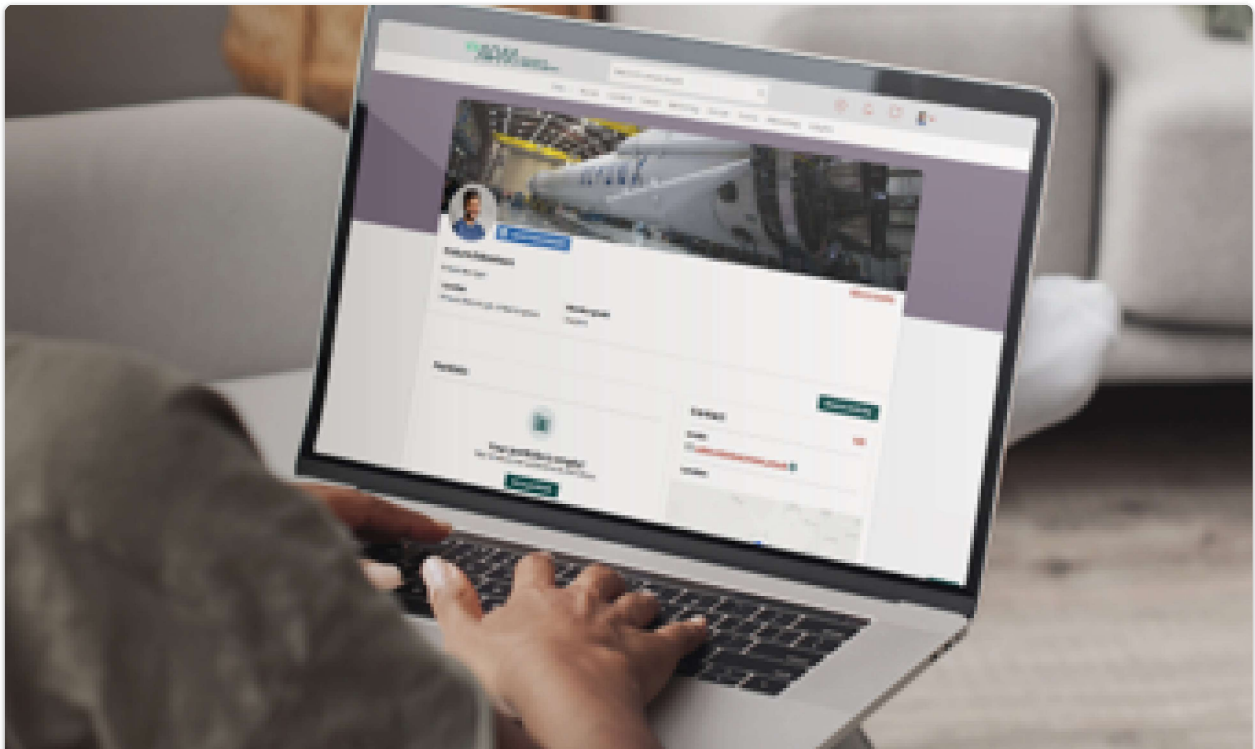
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